Name:_______________________ **Student ID:** _______________________

EGS3012S (2024) Test 1A

Total: 26 (MCQ)+10(Calc)=36 marks (~70s/question)

MCQ: Correct answer = +1, Incorrect = -0.25, Blank = 0 (Negative marking)

#	Question	Answer
1	*Fill in the blank: regions of low pressure on a constant height chart correspond to regions of ________ on a constant pressure chart:* ***(A)*** low pressure ***(B)*** low height ***(C)*** high pressure ***(D)*** high height	
2	*As the difference between the air temperature and wet-bulb temperature increases, the relative humidity:* ***(A)*** decreases ***(B)*** remains constant at a value less than 100% ***(C)*** increases ***(D)*** becomes undefined	
3	*At 20°C, an atmospheric parcel is observed to be saturated with water vapour. If the temperature of the parcel is increased to 35°C at constant pressure, with no addition of water vapour, the dew point temperature will be*: ***(A)*** 0°C ***(B)*** 15 °C ***(C)*** 20°C ***(D)*** 25°C ***(E)*** 55°C	
4	*If an air parcel is given a small push upward and it falls back to its original position, the atmosphere is said to be:* ***(A)*** stable ***(B)*** conditionally unstable ***(C)*** absolutely unstable ***(D)*** neutrally stable ***(E)*** saturated	
5	*The ozone layer is found in the:* ***(A)*** Troposphere ***(B)*** Stratosphere ***(C)*** Thermosphere ***(D)*** Mesosphere ***(E)*** Mesopause	
6	*If the environmental lapse rate is 0.01* $\mathrm{K \cdot m^{-1}}$ *and the surface air temperature is 25°C, then the air temperature at 500 m above the ground is:* ***(A)*** 25°C ***(B)*** 30°C ***(C)*** 20°C ***(D)*** 15°C	
7	*Supercooled cloud droplets are:* ***(A)*** ice crystals surrounded by air warmer than 0°C ***(B)*** liquid droplets that are cooler than the air around them ***(C)*** liquid droplets observed at temperatures below the freezing point of water ***(D)*** water droplets cooled to below 0K	
8	*Which one of the following processes releases latent heat?* ***(A)*** Evaporation ***(B)*** Melting ***(C)*** Condensation ***(D)*** Sublimation	

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9	*Which thermodynamic variable indicates the temperature at which human skin can be cooled to by evaporation?* ***(A)*** Wet-Bulb Temperature ***(B)*** Saturation vapour pressure ***(C)*** Dry-Bulb Temperature ***(D)*** Dew Point Temperature ***(E)*** Virtual temperature	
10	*Which of the following has the lowest albedo?* ***(A)*** Snow ***(B)*** Sand ***(C)*** Tar roads ***(D)*** Lakes ***(E)*** Grass	
11	*Which of these weather elements **always** decreases with height in the atmosphere?* ***(A)*** Virtual temperature ***(B)*** Environmental temperature ***(C)*** Air pressure ***(D)*** Relative humidity ***(E)*** Dew point temperature	
12	*Which of the following cloud types is a mid-level cloud?* ***(A)*** Cirrocumulus ***(B)*** Nimbostratus ***(C)*** Anvil ***(D)*** Altocumulus ***(E)*** Stratocumulus	
13	*Hail generally falls from which type of cloud?* ***(A)*** Cirrocumulus ***(B)*** Cumulonimbus ***(C)*** Brown Haze ***(D)*** Altocumulus ***(E)*** Stratocumulus	
14	*Compared to cold bodies, hotter bodies tend to emit radiation that has* ***(A)*** Higher energy and shorter wavelength ***(B)*** Lower energy and shorter wavelength ***(C)*** Lower energy and longer wavelength ***(D)*** Higher energy and longer wavelength	
15	*From an energy balance perspective, the atmospheric greenhouse effect influences global mean temperature primarily by* ***(A)*** Increasing albedo ***(B)*** Decreasing emissivity (***C)*** Increasing emissivity ***(D)*** Producing La Niñas ***(E)*** Decreasing albedo ***(F)*** Increasing insolation	
16	*In completely dry air, containing 0 water vapour, which of the following is **always** true* ***(A)*** Temperature and virtual temperature are equal ***(B)*** dew point temperature is greater than dry-bulb temperature ***(C)*** Wet bulb temperature is greater than virtual temperature ***(D)*** Saturated vapour pressure is not defined	

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17	*At the 200hPa pressure surface:* ***(A)*** geopotential height is about 20m greater than geometric height ***(B)*** geometric height is about 20m greater than geopotential height ***(C)*** geometric height is always exactly equal to geopotential height ***(D)*** geometric height is equal to geopotential height in saturated air	
18	*For air layers between 800 and 850hPa, which of the following would you expect would have the greatest thickness:* ***(A)*** average temperature = 20°C, average relative humidity = 5% ***(B)*** average temperature = 10°C, average relative humidity = 5% ***(C)*** average temperature = 20°C, average relative humidity = 95% ***(D)*** average temperature = 10°C, average relative humidity = 95%	
19	*As rain falls through warm air and is evaporated* ***(A)*** energy is released ***(B)*** the warm air cools ***(C)*** air pressure decreases ***(D)*** a warm front forms	
20	*If the absolute temperature of an object doubles, the maximum energy emitted goes up by a factor of* ***(A)*** 2 ***(B)*** 4 ***(C)*** 8 ***(D)*** 16 ***(E)*** 32	
21	*Along rural subtropical west coasts, which 2 cloud types are commonly seen:* ***(A)*** cumulonimbus, nimbostratus; ***(B)*** steam fog, advection fog; ***(C)*** brown haze, steam fog; ***(D)*** stratocumulus, cumulonimbus; ***(E)*** advection fog, stratocumulus	
22	*Among stable air columns with sea-level pressure of 1013hPa and the following surface properties, which would likely have the highest geopotential height at the 500hPa-level?* ***(A)*** Temperature = 30°C, Dew Point Temperature = 13°C ***(B)*** Temperature = 29°C, Dew Point Temperature = 17°C ***(C)*** Temperature = 35°C, Dew Point Temperature = 17°C ***(D)*** Temperature = 15°C, Dew Point Temperature = 12°C	
23	*Which of the following is a statement of the Clausius-Clapeyron relation?* ***(A)*** Saturation vapour pressure increases exponentially with temperature at a rate of about 7% per Kelvin ***(B)*** Density is proportional to pressure times temperature ***(C)*** Air parcel mixing ratios increase exponentially with temperature at a rate of about 7% per Kelvin ***(D)*** Air parcel saturated mixing ratios decrease exponentially with pressure at a rate of about 7% per Pascal	

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24	*If forced to rise by a mountain, an air parcel with which of the following properties would likely lead to lowest cloud base height?* ***(A)*** Temperature = 30°C, Dew Point Temperature = 11°C ***(B)*** Temperature = 29°C, Dew Point Temperature = 17°C ***(C)*** Temperature = 35°C, Dew Point Temperature = 26°C ***(D)*** Temperature = 15°C, Dew Point Temperature = 14°C ***(E)*** Temperature = 15°C, Dew Point Temperature = 11°C	
25	*If they were surrounded by the same fixed air mass, which of the following air parcels would tend to be least stable?* ***(A)*** Temperature = 31°C, Relative humidity = 25% ***(B)*** Temperature = 26°C, Relative humidity = 45% ***(C)*** Temperature = 15°C, Relative humidity = 50% ***(D)*** Temperature = 32°C, Relative humidity = 55% ***(E)*** Temperature = 14°C, Relative humidity = 26%	
26	*An air parcel experiencing deep convection–and thus rising up high into the troposphere–will tend to experience:* ***(A)*** Increasing temperature, decreasing pressure, increasing relative humidity ***(B)*** Decreasing temperature, decreasing pressure, increasing relative humidity ***(C)*** Increasing temperature, decreasing pressure, decreasing relative humidity ***(D)*** Decreasing temperature, increasing pressure, increasing relative humidity ***(E)*** Increasing temperature, increasing pressure, decreasing relative humidity	

Calculation questions:

(A) If the wavelength of maximum solar emission is 0.7×10^{-6}m, use Wien's Displacement Law $\lambda_{\max} = \frac{C}{T}$ (where $C = 2897 \times 10^{-6}\mathrm{m} \cdot \mathrm{K}$) to estimate the temperature of the sun's surface to the nearest Kelvin [3 marks].

(B) Assuming a dry adiabatic lapse rate of 0.01 $\mathrm{K} \cdot \mathrm{m}^{-1}$ and saturated adiabatic lapse rate of 0.006$\mathrm{K} \cdot \mathrm{m}^{-1}$, given a surface temperature of T = 20°C, and dew point temperature T_d = 16°C, for an air parcel raised adiabatically:

1. At roughly what height above the surface would you expect the cloud base to be found (answer to the closest 100m)? Write the answer only. [1 mark]
2. What would you expect the temperature to be to the nearest °C at a height of 1 km above the surface? Show your calculation and/or reasoning. [3 marks]
3. Estimate the pressure to the nearest Pascal at the 300m height if the surface pressure is 1013.00hPa and the intervening layer is neutrally stable. Show your calculation and/or reasoning [3 marks]

Answer Calculation questions on the next sheet:

Name:________________________ Student ID: ________________________

Calculation answers:

A: formula used: ________________________

substitution: ________________________

answer: ________________________

B1: ________________________

B2: formula used: ________________________

substitution and/or conversions: ________________________

____

answer: ________________________

B3: formula used: ________________________

substitution and conversions: ________________________

____

answer: ________________________

See formula sheet on the next page: